A Gentle Introduction to Weyl Symbols

1 The main idea

A differential operator acts on a function. A *symbol* is a way to describe that operator using an ordinary function of position and wavenumber. It is useful because many questions about an operator—what waves it supports, how it propagates them, and how two operators combine—become easier to see in this position–wavenumber picture.

For a wave $u(x) = e^{i\xi x}$, the derivative measures its wavenumber:

$$Du = \xi u, \qquad D = -i\partial_x. \tag{1}$$

So D is represented by the variable ξ. For example, the constant coefficient operator $D^2 + k^2$ has symbol $\xi^2 + k^2$. Its symbol immediately shows which wavenumbers make $(D^2 + k^2)u = 0$.

When coefficients vary with position, position and wavenumber both matter. The operator $a(x)D$ has a symbol involving both $a(x)$ and ξ. There is a small complication: multiplication by $a(x)$ and differentiation do not commute. Different choices for their ordering give slightly different symbols. Weyl's choice treats position and wavenumber symmetrically.

2 Weyl quantization

Given a function $p(x, \xi)$, its Weyl operator is

$$(\mathrm{Op}^w(p)u)(x) = \frac{1}{2\pi} \iint e^{i(x-y)\xi} p\left(\frac{x+y}{2}, \xi\right) u(y)\, dy\, d\xi. \tag{2}$$

The key feature is the midpoint $(x + y)/2$: the coefficient is sampled halfway between where the input and output are evaluated. This makes Weyl quantization symmetric in position and wavenumber.

Some basic examples are

$$\mathrm{Op}^w(a(x))u = a(x)u, \tag{3}$$

$$\mathrm{Op}^w(\xi)u = Du = -iu_x, \tag{4}$$

$$\mathrm{Op}^w(a(x)\xi)u = \frac{1}{2}\big(aD + Da\big)u = -i\left(au_x + \frac{1}{2}a_x u\right). \tag{5}$$

Thus the symbol $a(x)\xi$ represents the *symmetrically ordered* operator, not simply aD. In fact,

$$aD = \mathrm{Op}^w\left(a(x)\xi + \frac{i}{2}a_x(x)\right). \tag{6}$$

The extra term records the effect of the ordering. For real p, the Weyl operator is symmetric (subject to the usual domain and boundary conditions).

3 Combining symbols

Applying one operator after another corresponds to combining their symbols with a product called the *Weyl product*, written $p\#q$. Its first terms are

$$p\#q = pq + \frac{i}{2}\left(p_x q_\xi - p_\xi q_x\right) - \frac{1}{8}\left(p_{xx}q_{\xi\xi} - 2p_{x\xi}q_{\xi x} + p_{\xi\xi}q_{xx}\right) + \cdots . \tag{7}$$

The first term is ordinary multiplication. The remaining terms correct for the fact that position-dependent coefficients and derivatives do not commute. If the coefficients vary slowly compared with a wavelength, these corrections are often small, so ordinary multiplication gives a useful first approximation.

For example, the symbol of a commutator is, to leading order,

$$\sigma\big([\mathrm{Op}^w(p), \mathrm{Op}^w(q)]\big) \simeq i\left(p_x q_\xi - p_\xi q_x\right). \tag{8}$$

This expression is the phase-space Poisson bracket (with the convention shown here). It tells us that operator order matters most where the two symbols vary across phase space.

4 A variable-coefficient wave operator

Consider the one-dimensional vertical operator

$$B = \partial_x\big(g(x)\partial_x\big) + V(x), \qquad g(x) > 0. \tag{9}$$

Since $D = -i\partial_x$, this is $B = -DgD + V$. Its exact Weyl symbol is

$$b(x,\xi) = -g(x)\xi^2 + V(x) - \frac{1}{4}g_{xx}(x). \tag{10}$$

The leading part $-g\xi^2 + V$ is the local dispersion relation. The smaller $-g_{xx}/4$ term is a correction caused by the variation of the coefficient and by symmetric operator ordering. In a slowly varying medium, the principal symbol often explains the main wave behavior, while the lower order terms describe corrections to phase and amplitude.

In terrain-following Helmholtz analysis, the same idea applies with position η and vertical wavenumber q. The symbol of the vertical operator gives the local dispersion relation at each range, while its range variation couples forward and backward waves. Weyl notation keeps the operator ordering visible and provides a systematic way to calculate those coupling and higher-order terms.

Convention. This note uses $D = -i\partial_x$, the kernel $e^{i(x-y)\xi}$, and the Poisson bracket $\{p,q\} = p_x q_\xi - p_\xi q_x$. Changing the Fourier or bracket convention changes some signs together; the underlying operator statements are unchanged.